Suman Thapa

@ sumanthapa3271@gmail.com | LinkedIn | GitHub | Website | Lemgo, Germany

Education

Technische Hochschule Ostwestfalen-Lippe — Lemgo, Germany
M.SC. Information Technology — *2024 – Present*

London Metropolitan University (in collaboration with Islington College) — Kathmandu, Nepal
BSc (Hons) Computing ***(First Class Honours)*** — *2019 – 2022*

Relevant Experience

SAP SE — Walldorf, Germany
SAP Full Stack Developer (Work-Study) — *Aug 2024 – Present*

- Developed full-stack enterprise applications using **SAP CAP (Node.js)** and **SAPUI5**, building scalable **OData V4** services with custom business logic and annotations to integrate with **SAP HANA** and external APIs.
- Implemented an integration with OpenAI's **Large Language Models (LLMs)**, utilizing **function calling** to parse natural language queries into structured data for triggering backend **CAP services**.
- Secured applications using Authorization and Trust Management **(XSUAA)** and configured **SAP BTP Destinations** to ensure safe and reliable service-to-service communication.
- Wrote unit tests for **CAP** services to ensure reliability.
- Managed **Jenkins** CI/CD pipelines and deployed applications to the **SAP BTP**, Cloud Foundry environment.

Innovate Tech — Kathmandu, Nepal
Associate Backend Developer — *Jan 2023 – Mar 2024*

- Worked on scalable **microservices** to develop features for a digital learning platform, *My Second Teacher*.
- Developed and documented RESTful **APIs** using **Express** and **Koa** frameworks.
- Built a real-time chat system using **socket.io** with private messaging, group chat, and user presence tracking for thousands of concurrent users.
- Utilized the **mongoose** library to interact with a **MongoDB** NoSQL database, including defining schema models, performing CRUD operations, and implementing queries for data retrieval.
- Implemented the **Sequelize** ORM to communicate with **PostgreSQL**.
- Integrated **AWS** services including **Lambda, SQS, and SNS** to develop decoupled and reusable application features, such as asynchronous notifications.
- Conducted in-depth research on **Prosody XMPP server** to evaluate its feasibility for enhancing real-time communication features.

Vairav Technology — Kathmandu, Nepal
Full stack software engineer — *May 2022 – Dec 2022*

- Contributed to the development of **Vairav Trident**, an AI-enabled **SIEM** solution with built-in detection capabilities and integrated **threat intelligence**, ensuring comprehensive security visibility and continuous threat mitigation.
- Developed scalable **microservices** that communicate over **gRPC** to process high-volume security event logs.
- Developed **APIs** using **Apollo GraphQL** to serve data for security visualization dashboard and generating reports.
- Implemented a trigger based **CDC (Change Data Capture) / Audit** system in **PostgreSQL** to maintain real-time audit trail of security logs.
- Developed an interactive frontend using **React.js** and **Elastic UI**, featuring visualization of security events.
- Implemented role based **authentication/authorization middleware** on backend services using **keycloak** as JWT provider.
- Utilized **Prisma ORM** model to communicate with PostgreSQL and the **Elasticsearch** client library for querying and indexing security logs, ensuring efficient data manipulation across SQL and NoSQL databases.

Level 9 Tech — Kathmandu, Nepal
Backend Developer Intern — *Aug 2021 – Nov 2021*

- Built **REST API** endpoints for a learning management system and airline booking platform using **Django Rest Framework** and **PostgreSQL**.
- Wrote **unit tests** for API endpoints to ensure code quality.

Skills

Programming Languages: TypeScript, JavaScript, Python, SQL.

Frontend: React, SAP UI5.

Backend: Node.js, Express.js, Koa, SAP CAP, GraphQL, REST API, gRPC.

Databases: PostgreSQL, MongoDB, SAP HANA, Elasticsearch.

Cloud & DevOps: AWS, SAP BTP, Docker, Jenkins, Git.

Monitoring & Analytics: ELK stack, Grafana.